



CREDIT EDA ASSIGNMENT

Submitted By Biplab Mondal
(Batch-C54)

PROBLEM STATEMENT

- Bank Loan company need to find out how to Approve or Reject the Loan application forms.
- To whom Bank should provide the Loans.
- When the company receives a loan application, the company has to decide for loan approval based on the applicant's profile.
- There are two steps of risk factors with Bank's decision.
 - If the applicant is likely to repay the loan, then not approving the loan results in a loss of business to the company.
 - If the applicant is not likely to repay the loan, i.e. he/she is likely to default, then approving the loan may lead to a financial loss for the company.

CREATE A NEW JUPYTER NOTEBOOK

- Import all the necessary python libraries.



READ THE DATASETS

- First, We need to read the three datasets given the Bank.
 - **Application Data** : Contains all the information of the client at the time of application. The data is about whether a client has payment difficulties.
 - **Previous Application Data** : Contains information about the client's previous loan data. It contains the data on whether the previous application had been Approved, Cancelled, Refused or Unused offer.
 - **Columns Description** : Data dictionary which describes the meaning of the variables.

DATA CLEANING

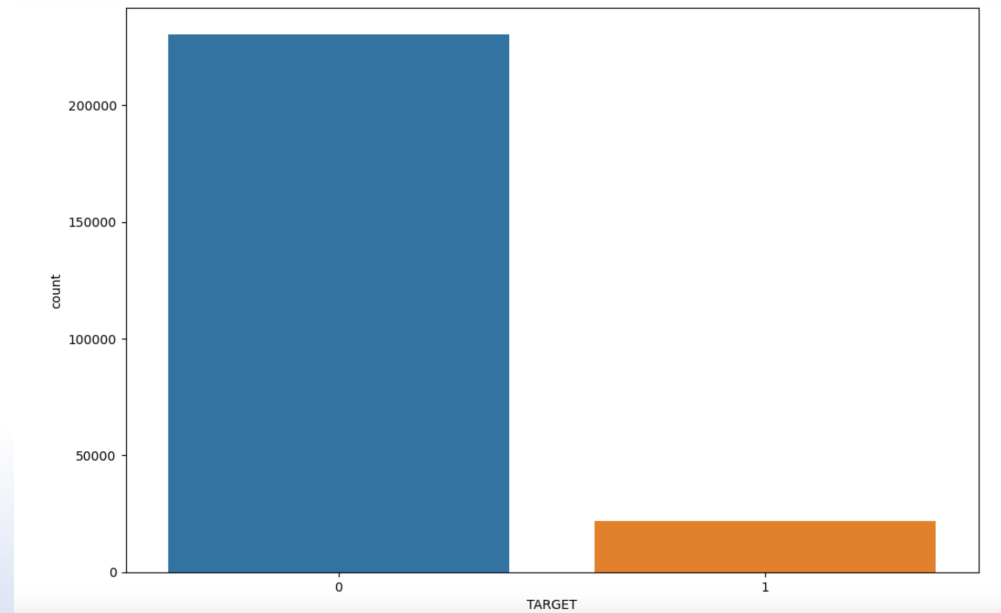
- Fix the Missing Values.
 - Drop the Columns those are having missing values > 40 %
 - Fill the rest of the columns with mean, median, and mode.
 - Impute the values in continuous and categorical missing value columns.
- Fixing invalid data types.
- Fixing Negative values where its required (Date, Age, etc.)
- Checking Outliers and fixing if required.

ANALYSIS

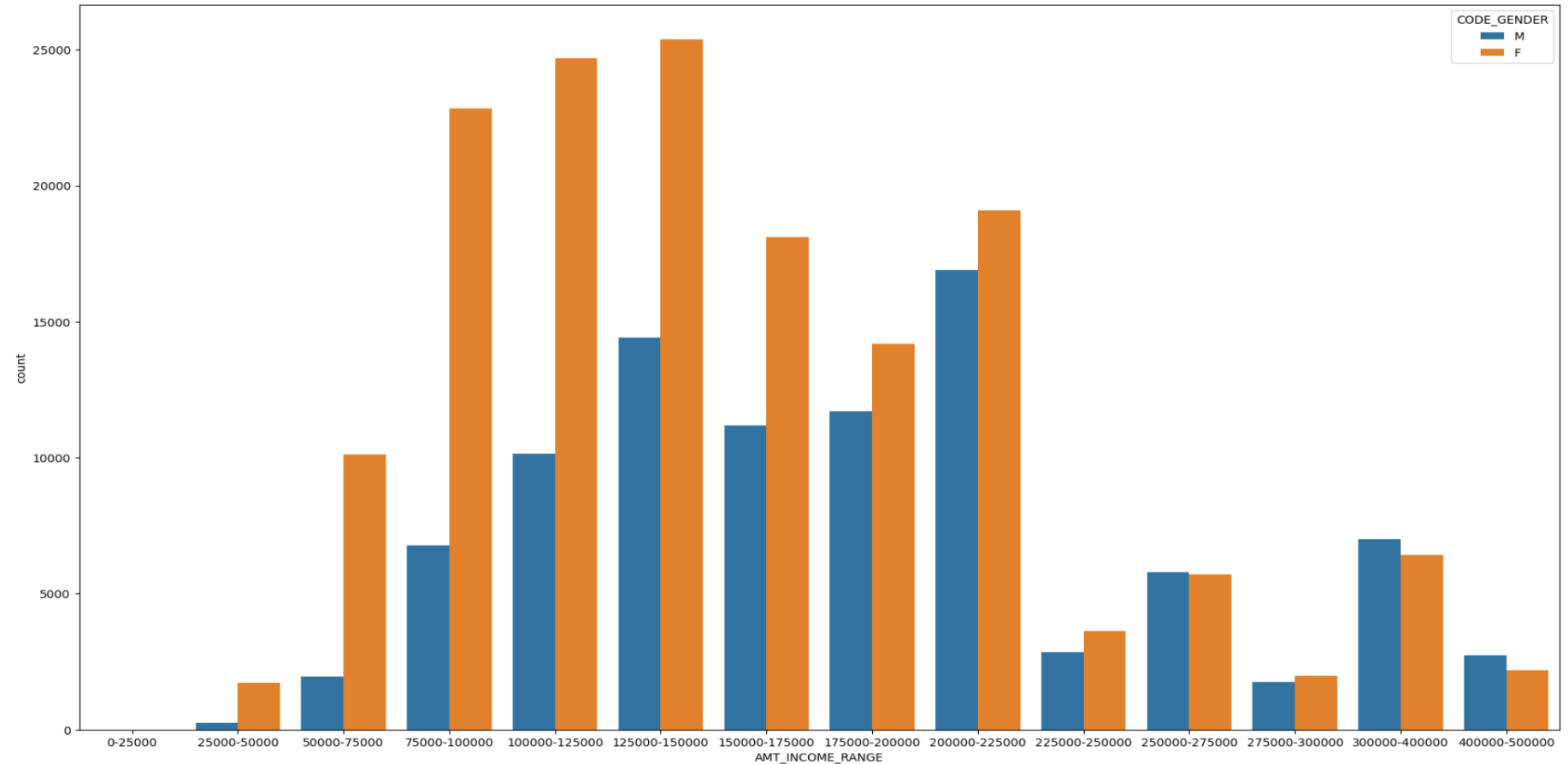
- Binning the categorical columns.
- Plotting and Visualizing.
- Checking, analyzing, and removing unwanted columns those are not required for the analysis.
- Checking ratio for the Data imbalance.
- Performing Univariate, Bivariate, and Multivariate analysis for the required variables.
- Finding Correlations between variables.
- Merge Application_Data and Previous_Application_Data by their common ID and performing all three types of analysis (Univariate, Bivariate, and Multivariate).

IMBALANCE RATIO FOR TARGET VARIABLE

Application Dataset Target variable's imbalance ratio (10.55) shows the less default applicants are there in the data.

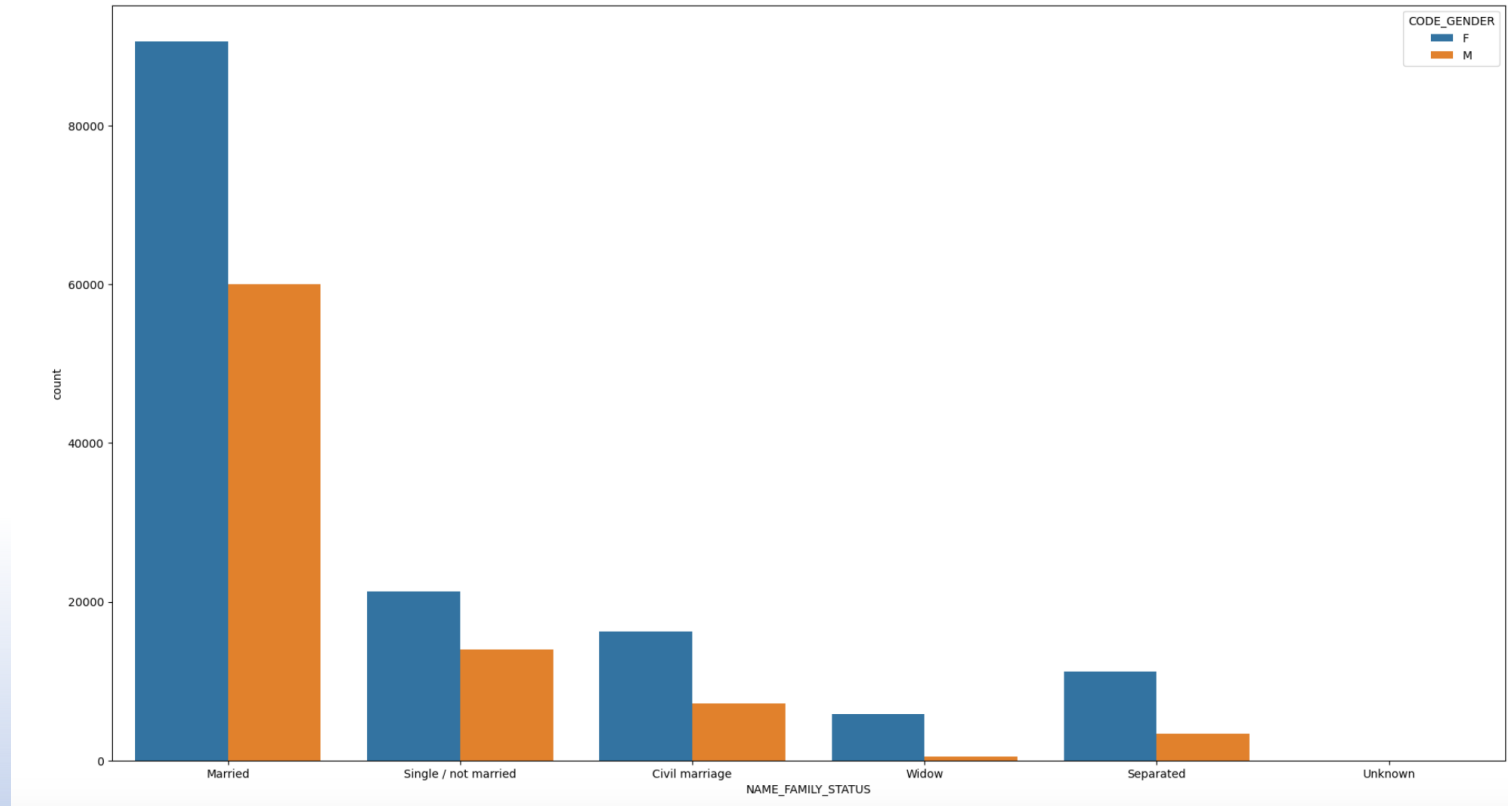


UNIVARIATE ANALYSIS



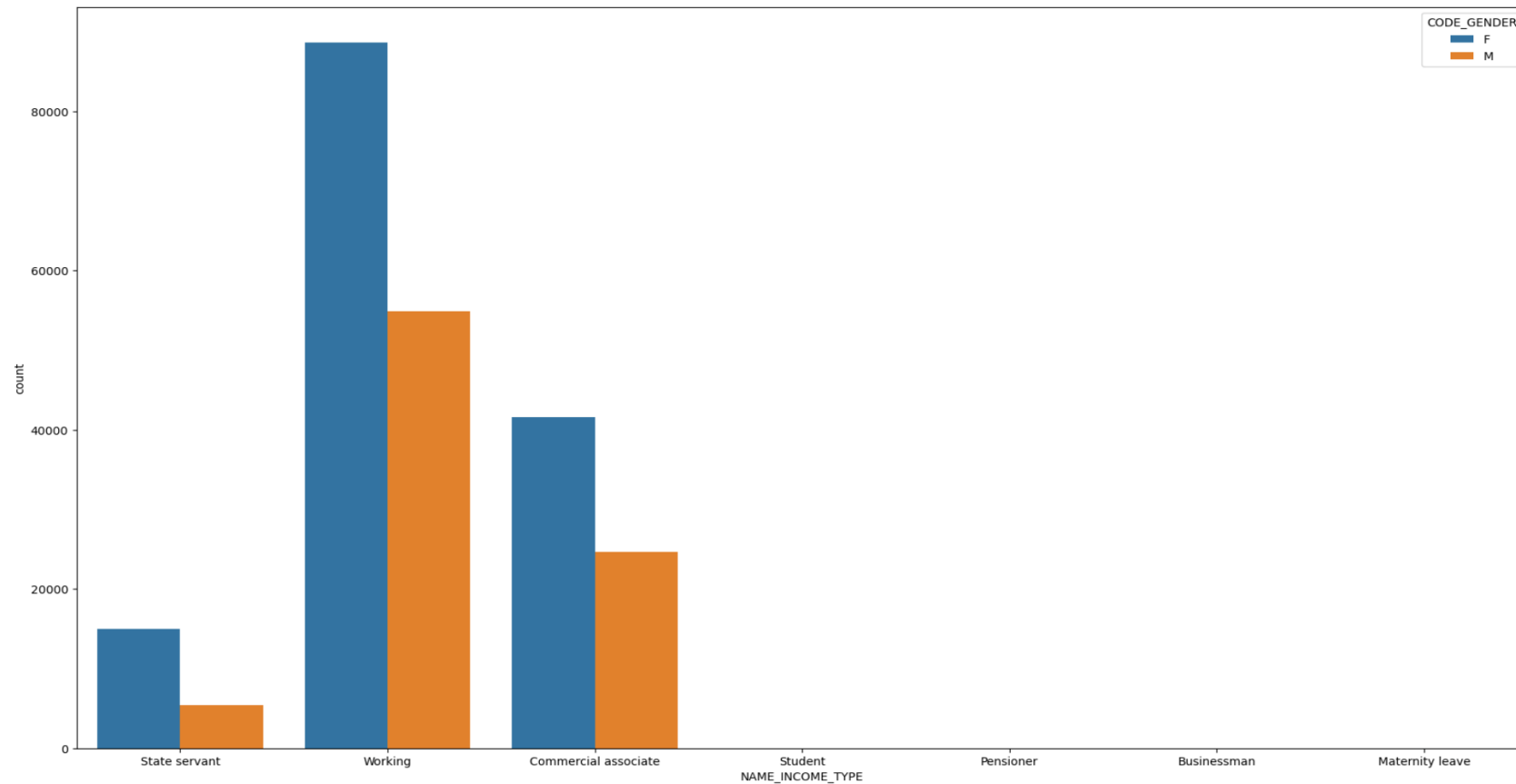
Income range between 75000 to 225000 having higher applicants. Females are more aggressive to take loans. Targeted people would be under 75000 to 225000 as shown in plot.

UNIVARIATE ANALYSIS



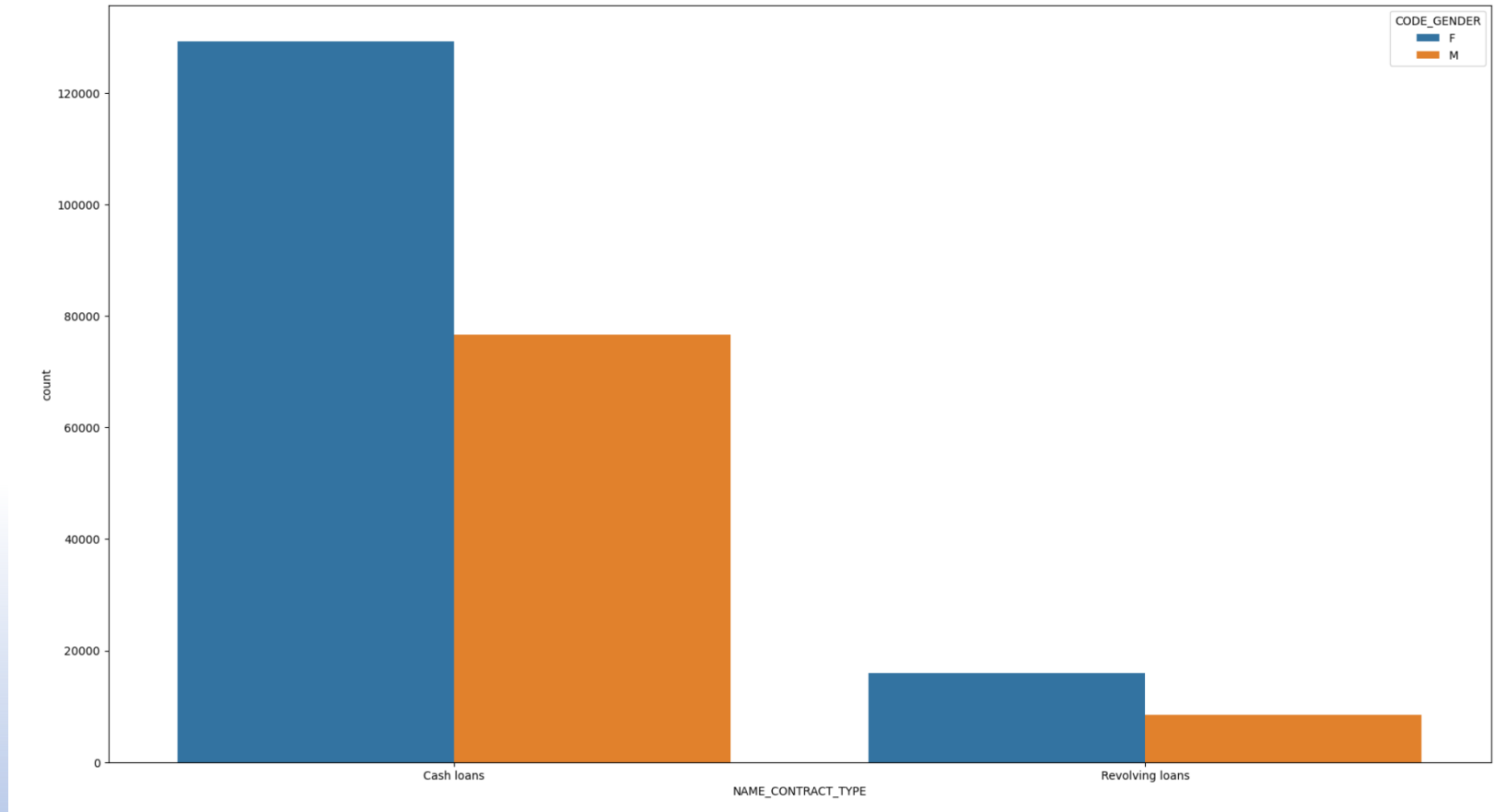
Married applicants are more in TARGET_0 category i.e. Non-defaulters. Bank can focus on these applicants. "Married-Female" category applicants are the topmost to accept their applications.

UNIVARIATE ANALYSIS



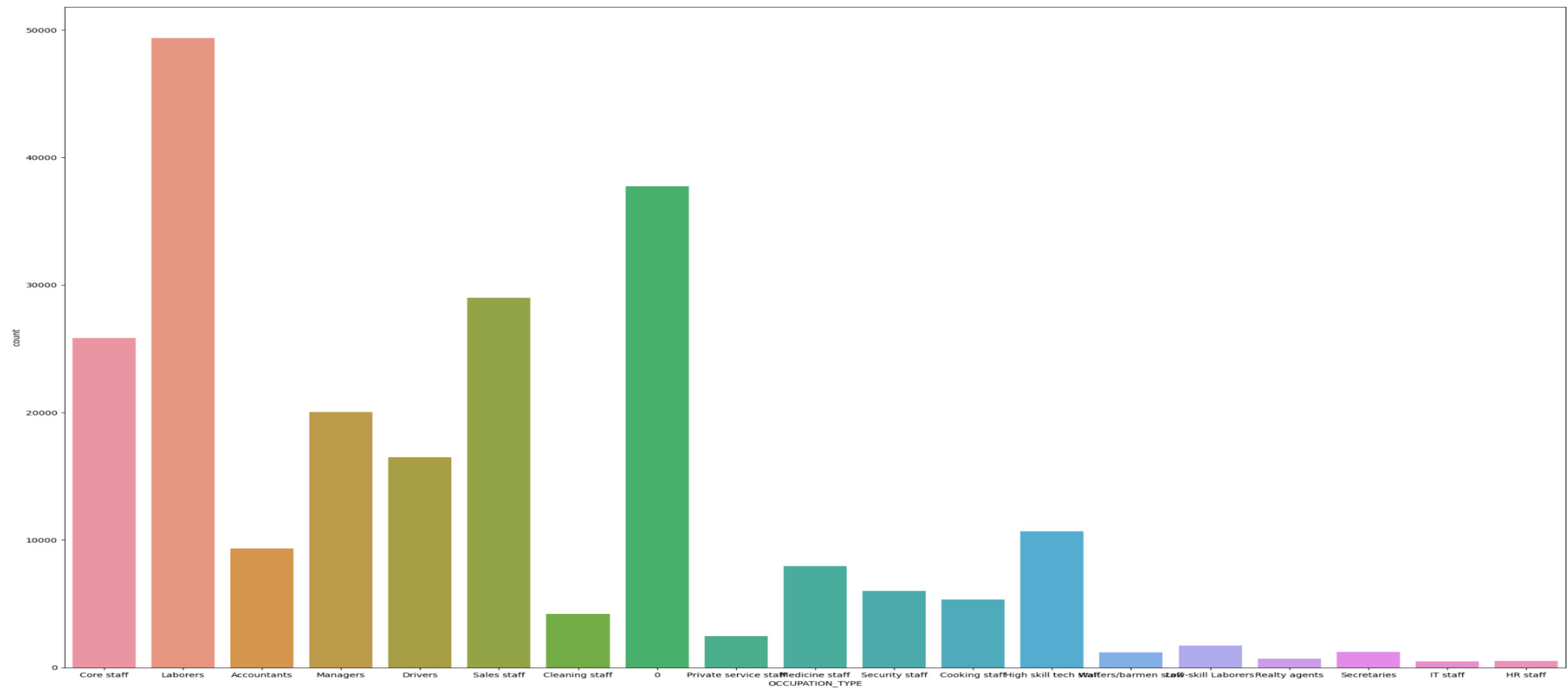
"Female Working" people are the best for loan approval as they have highest number of on-time payment. Apart from "State servant", "Working people", "Commercial associate", others are not that much focusable for non-defaulters.

UNIVARIATE ANALYSIS



“Cash loans” are higher number in credits with no defaults.

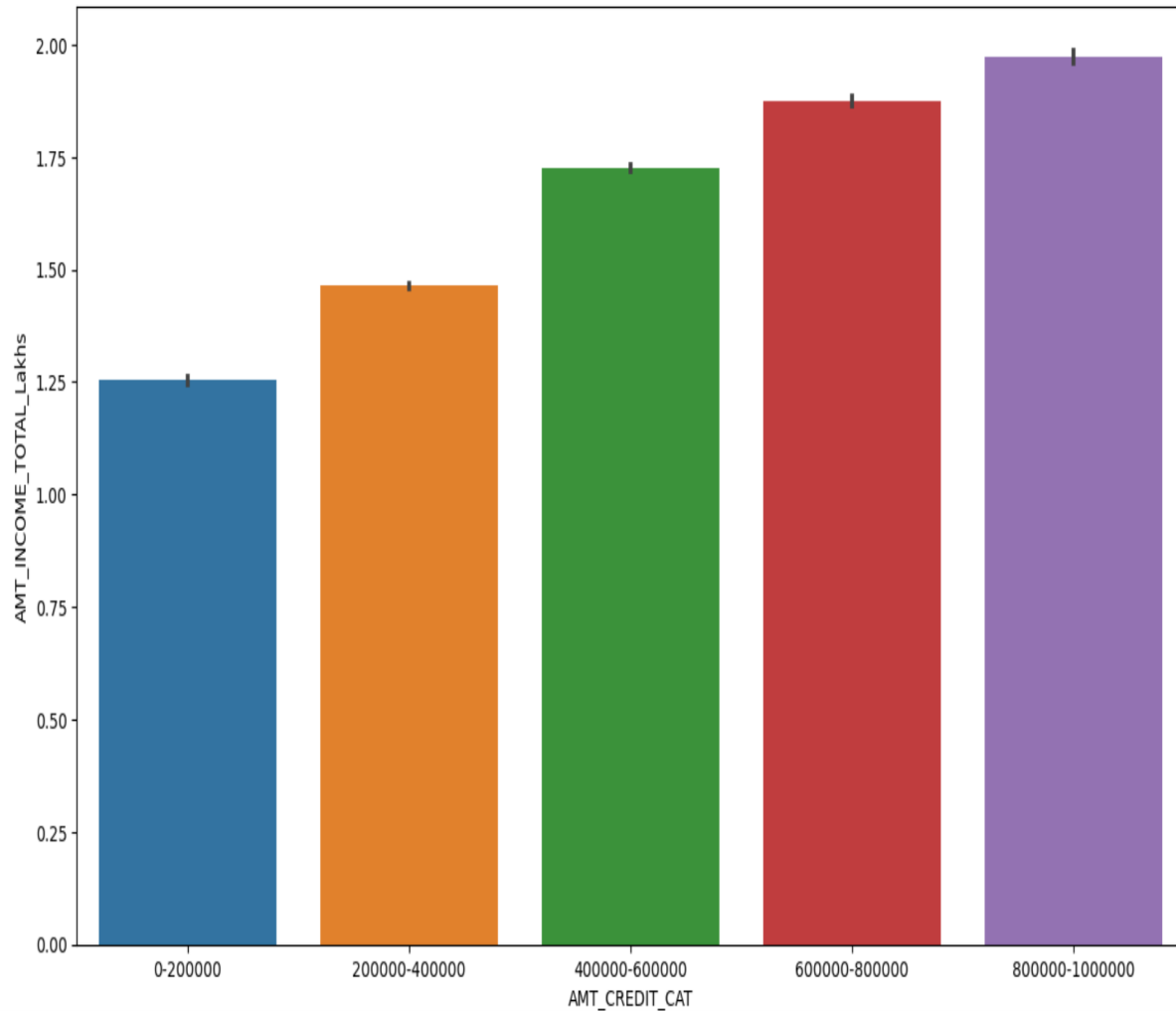
UNIVARIATE ANALYSIS



“Labour” occupied people could be the most targeted for Loans as less defaulters.

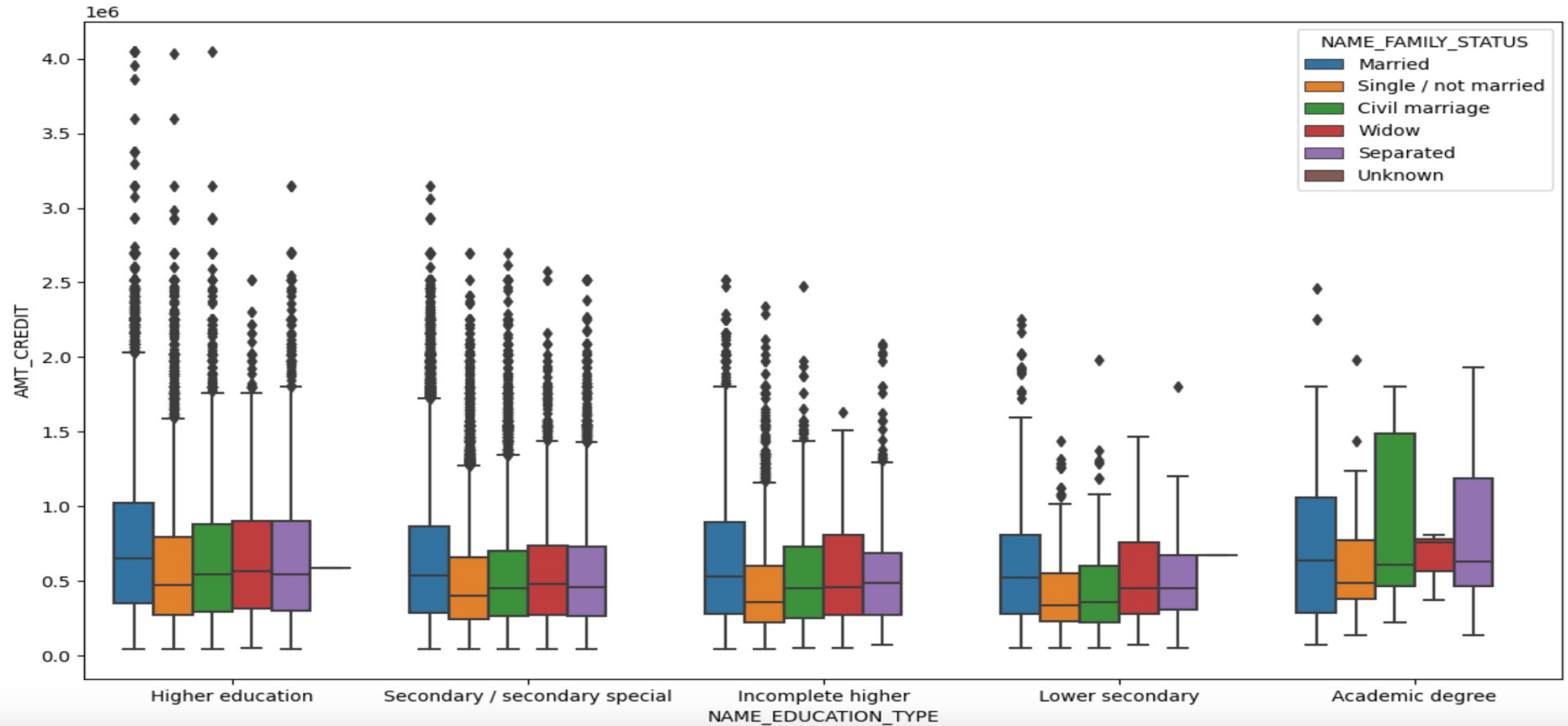
BIVARIATE ANALYSIS

Higher income applicants are having higher credit amounts with less defaulters.



MULTIVARIATE ANALYSIS

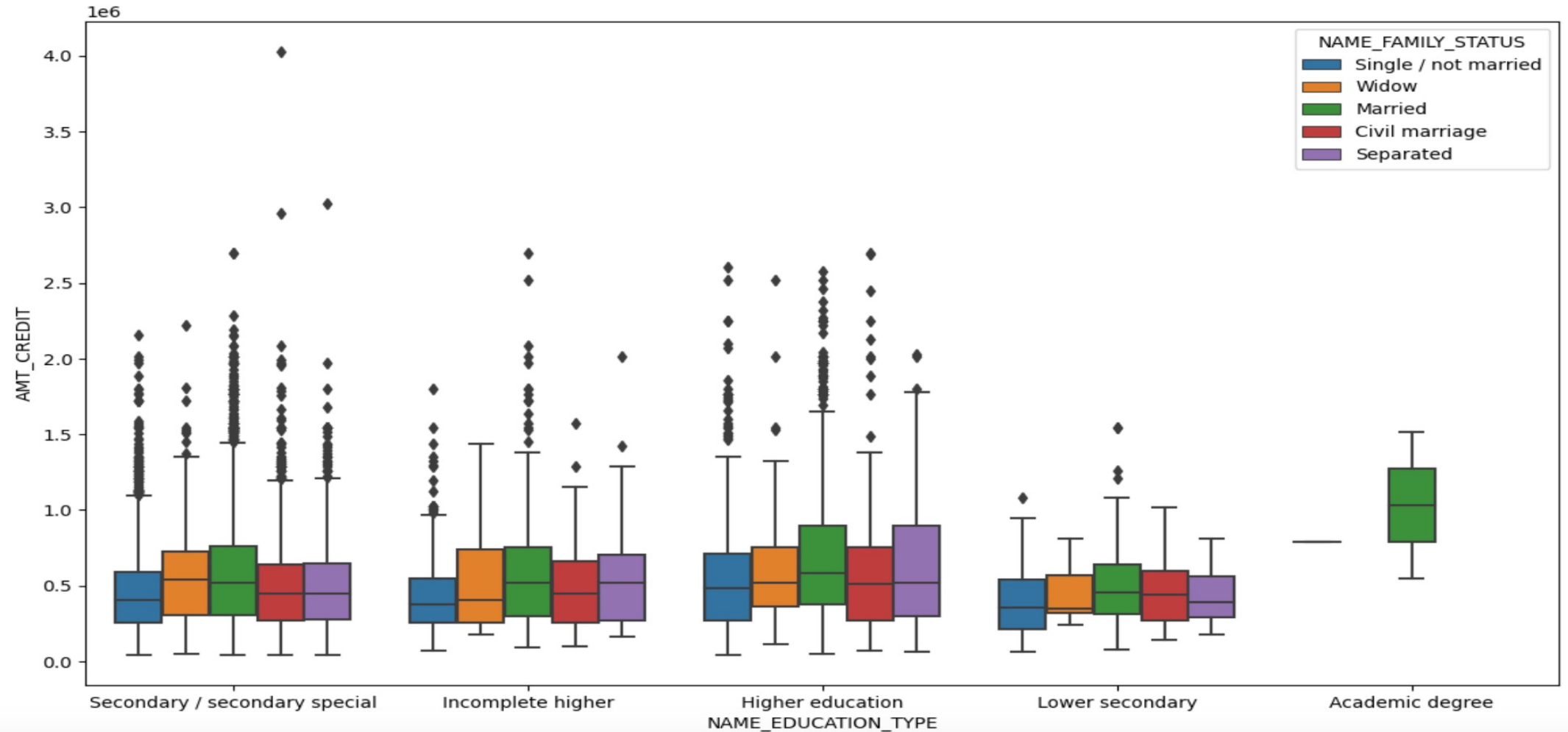
Credit Amount vs Education Status (Non-defaultties)



Academic degree holder with Civil marriage, and separated peoples are more targeted.

MULTIVARIATE ANALYSIS

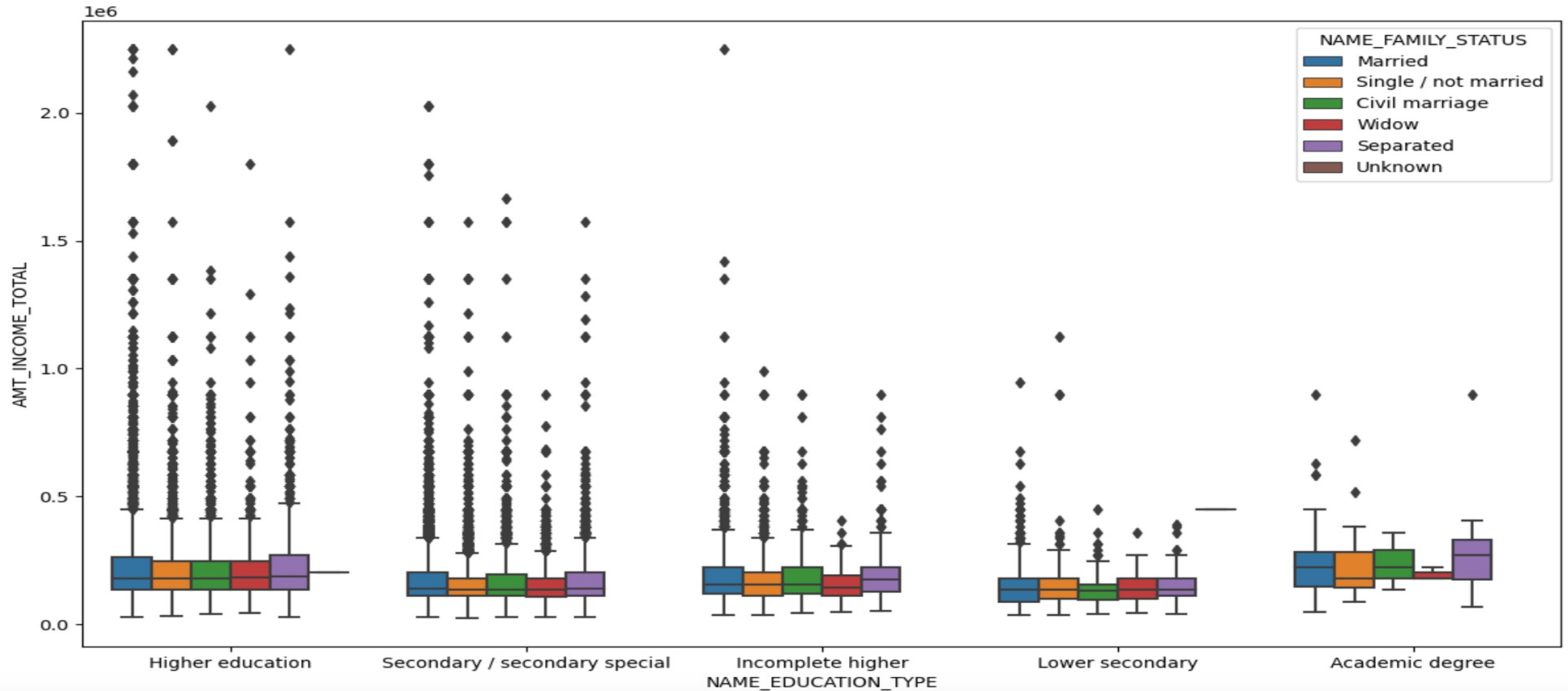
Credit Amount vs Education Status (Defaulties)



Academic degree "Married" peoples are more defaulters

MULTIVARIATE ANALYSIS

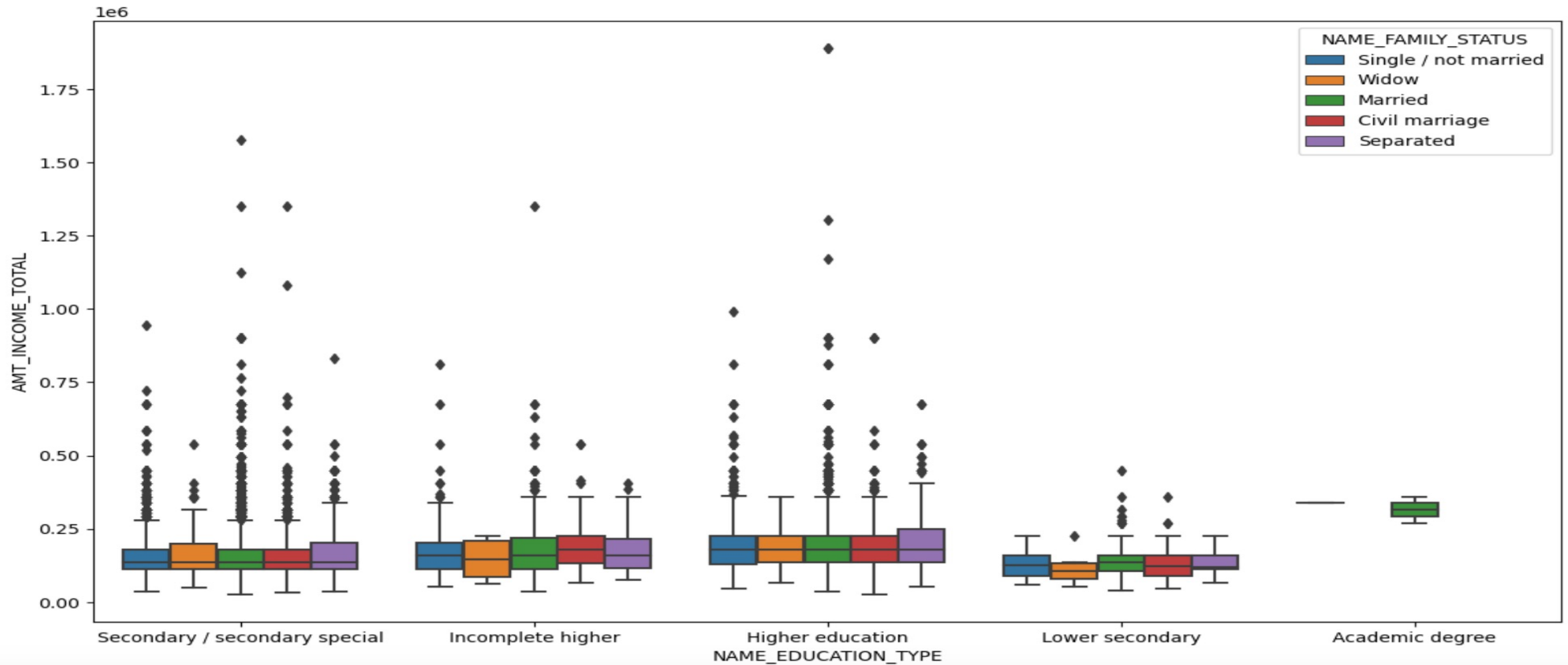
Total Income vs Education Status (Non-defaulties)



Higher Education peoples are more non-defaulters

MULTIVARIATE ANALYSIS

Total Income vs Education Status (Defaulties)



Academic degree with married peoples are more in defaulters.

CONCLUSIONS

Bank should target the applicants who are “under 30 to 50 years of age”.

“Females” are more in numbers to give loans as they are higher in non-default percentages.

“Business Entity Type 3” and “Self-employed” persons are more acceptable applicants.

"Very low" and "Low" income category people can be targetable.

Non-defaulters are 10 times more than Defaulters as the Data imbalance ratio is 10.55.

Income range between 75,000 to 1,50,000 having higher applicants. Females are more aggressive to take loans with less defaults.

Married Applicants, especially female married applicants are more in non-defaulters.

"Female Working" people are the best for loan approval as they have highest number of on-time payment.

Apart from "State servant", "Working people", "Commercial associate", other income type categories are not that much focusable for non-defaulters.

“Cash loans” are higher number in credits with no defaults.

“Labour” occupied people could be targeted for Loans.

“Secondary and Secondary Special education” applicants could be targetable.

Bank should provide loans to “House Apartments” and “Office Apartments” peoples as their default risk is less.

THANK YOU

